Digital Twin Anchored Generation: Adaptive Querying and Phenotypic Refinement

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***Abstract*—Digital Twin Anchored Generation (DTAG) couples a language interface with Large Science Models (LSMs) of survey-response dependencies. Conditional predictions for unasked items define a *predictive phenotype*, refined by successive answers; geometric adaptive querying selects questions by expected movement of a pole-relative index. We distinguish such inference from *phenotypic construction*, in which interaction changes future reports or beliefs. Native GSS 2024 simulations with six pinned personas yielded 960 adaptive 20-question trajectories and 2,880 fixed-item order replays. Reordering changed coupled sample paths, including some endpoint signs, but all 54 signed-mean order-contrast bootstrap confidence intervals contained zero. These experiments evaluate the native conditional model, not held-out human refinement accuracy, language-mapping accuracy, or persuasion. Future randomized human studies will test response prediction, directional adaptive querying, and order effects with independent outcomes and delayed follow-up.**

***Index Terms*—Computational theory of mind, opinion dynamics, large science models, digital twins, survey methodology, social influence, polarization, latent geometry, adaptive questioning.**

I. Introduction

Belief systems comprise dependencies among economic, religious, institutional, and political attitudes, not only individual responses. Classical studies of issue constraint [1], [2] and later work on partisan sorting and alignment [3], [4] motivate this high-dimensional view. We ask whether sequential questioning reveals or helps construct such a system.

DTAG links natural-language queries to an LSM of empirical conditional survey responses. Each answer anchors a predictive phenotype, updates predictions for unasked questions, and moves a pole-relative geometric index. Adaptive queries select items by expected displacement, making sequential behavior measurable rather than merely plausible prose.

Synthetic ideological movement is not persuasion. We formalize *passive refinement* of fixed beliefs versus *reflexive construction* through interaction, then specify human tests of whether learned conditional geometry predicts more than inference.

Working research draft, October 2026. University of Kentucky. DTAG computations and human-subject studies described as proposed must not be interpreted as observed results.

II. Prior Work and the Scientific Gap

A. Belief systems, cognitive organization, and survey response

Converse [1] identified weak issue constraint outside politically sophisticated groups; Zaller [2] modeled reports as samples from accessible considerations. Later studies separate partisan sorting from issue alignment [3] and show how homophily and influence couple otherwise weak affinities [4]. These motivate measuring conditional attitude organization rather than assuming a single axis.

Consistency pressures [5], survey wording and order [6], and motivated evaluation [7] can change reports. Temporary salience, scale artifacts, reconstruction, and lasting revision require separate tests of persistence and generalization.

B. Quantitative influence and experimental evidence

DeGroot [8], Friedkin–Johnsen [9], and bounded-confidence [10] models specify social transitions but typically use low-dimensional individual states. DTAG instead models high-dimensional conditional structure without yet identifying a causal human transition operator.

Bail *et al.* [11] randomized incentives to follow Twitter bots providing cross-cutting political content. Republicans assigned a liberal bot later reported more conservative views; the corresponding Democratic estimate was not significant. This asymmetry motivates, but does not validate, DTAG. Campaign-contact experiments found small general-election mean effects [12], whereas some structured conversations produced more persistent reported changes [13]. Prediction must address heterogeneity and persistence.

C. Theory of mind and generative simulation

Theory of mind infers unobserved mental states from behavior [14], [15]; inverse planning formalizes this inference [16]. DTAG tests a narrower, falsifiable task: predict distributions of *unasked attitudes* from observed answers. Synthetic LLM personas [17], [18] may reproduce profiles without calibrated individual conditionals; DTAG exposes states, probabilities, and response provenance.

III. Large Science Models and the Predictive Phenotype

A. Conditional dependency model

Let $X = (X_1, \ldots, X_n)$ range over $\mathcal{X} = \prod_i \mathcal{A}_i$, with finite response alphabets $\mathcal{A}_i$ and survey population, wave, or context c. An LSM is the conditional family

$$\Phi_c = \{\phi_{i,c}(a \mid x_{-i}) : i = 1, \ldots, n\}, \tag{1}$$

estimated using coupled conditional inference trees [19]–[21]. Edges $j \to i$ indicate prediction, not causation; separately fitted conditionals may be jointly incompatible [22].

The respondent state is a partial assignment $s = \{X_j = a_j : j \in S\}$. In $\widetilde{x}_{-v}(s)$, all unanchored coordinates remain missing (no modal completion); the native trees marginalize missing splits by subtree-mass-weighted branch mixtures. This is a tree-wise prediction rule, not necessarily marginalization under one coherent joint law. For unasked $v \notin S$,

$$p_\Phi(a \mid s, v) = \phi_{v,c}(a \mid \widetilde{x}_{-v}(s)). \tag{2}$$

The vector

$$\mathcal{B}_\Phi(s) = \big(p_\Phi(\cdot \mid s, v)\big)_{v \notin S} \tag{3}$$

is the *predictive phenotype*: conditional distributions and uncertainty for unasked responses.

B. Geometry as predictive equivalence

Conditional predictions induce a distance between complete or partial states x, y:

$$\theta_\Phi(x, y) = \frac{1}{n} \sum_{i=1}^{n} \sqrt{D_{\mathrm{JS}}(\phi_{i,c}(\cdot \mid x_{-i}), \phi_{i,c}(\cdot \mid y_{-i}))}. \tag{4}$$

Here D_{JS} is Jensen–Shannon divergence and the native code evaluates *qdistance*. Predictively indistinguishable states may be at zero distance: this is generally a pseudometric of *predictive consequence*, not a postulated Euclidean manifold of minds.

With 80% responses masked [21], mean LSM-distance distortion declined about 54.9% (GSS) and 60.9% (Eurobarometer). These are geometric reconstruction gains, not exact-label accuracy or causal influence.

For GSS models with registered left and right reference states s_L, s_R, DTAG reports

$$I_\Phi(s) = \frac{\theta_\Phi(s_L, s) - \theta_\Phi(s_R, s)}{\theta_\Phi(s_L, s_R)}. \tag{5}$$

The semantically defined poles [21] specify 27 opposing GSS 2024 responses on social policy, civil liberties, public spending, and religion (Table I); all other coordinates remain unspecified. They are reference configurations, not typical respondents. The pole-relative index is not a self-reported ideology score.

IV. DTAG: A Conditional Theory-of-Mind Interface

A. Architecture and generative semantics

A semantic interface maps question q to survey variables $M(q)$. The LSM predicts responses, the answer anchors the state, and an LLM verbalizes it. For draw mode,

$$A_t \sim p_\Phi(\cdot \mid s_t, v_t), \qquad v_t \in M(q_t), \tag{6}$$

$$s_{t+1} = s_t \cup \{v_t = A_t\}, \tag{7}$$

$$\mathcal{B}_{t+1} = \mathcal{B}_\Phi(s_{t+1}). \tag{8}$$

In *draw mode*, an answer is sampled, not chosen by the optimizer; *pinned* anchors remain fixed. The system also supports modal answering, validated/multi-item mapping, and finite-memory *eviction* of old anchors, none used in the reported native runs. The fixed LSM conditions responses; it does not identify persuasion.

Figure 1 combines the architecture and native experiment.

B. Sequential geometry and adaptive questioning

For every candidate item v and response a, define

$$\Delta_v(a; s) = I_\Phi(s \cup \{v = a\}) - I_\Phi(s), \tag{9}$$

$$\mu_v(s) = \sum_{a \in \mathcal{A}_v} p_\Phi(a \mid s, v) \Delta_v(a; s), \tag{10}$$

$$\sigma_v^2(s) = \sum_{a} p_\Phi(a \mid s, v)(\Delta_v(a; s) - \mu_v(s))^2. \tag{11}$$

Adaptive policies optimize $\mu_v(s)$ under conditional sampling; modal-response objectives instead use the most probable answer. A modal sequence is not a sampled respondent population. These displacements are model-relative, not proven human interventions.

The predictive change at unasked item j after observing (v, a) is

$$R_j(s; v, a) = D_{\mathrm{JS}}\big(p_\Phi(\cdot \mid s, j), p_\Phi(\cdot \mid s \cup \{v = a\}, j)\big). \tag{12}$$

A weighted score $\mathcal{P}_\Phi(s; v, a) = \sum_{j \notin S \cup \{v\}} w_j R_j(s; v, a)$, with prespecified $w_j \geq 0$, quantifies predictive redistribution, not causal propagation.

C. Difference from RAG and language-model personas

RAG provides grounding text [23]; profile-prompted LLM personas seek plausible or distributionally faithful responses [17], [18]. DTAG instead exposes a native conditional model whose predictions for unasked responses can be calibrated as anchors accumulate. It complements, rather than replaces, text retrieval.

V. The Interpretation of Sequential Questions

A. Phenotypic refinement: observing a fixed system

Under passive refinement, respondents possess fixed phenotypes $X^\star$ and answers increasingly constrain predictions along $s_0 \subset s_1 \subset \cdots \subset s_T$. For target items Y, the information value of query v is

$$\mathcal{G}(v \mid s) = H(Y \mid s) - \mathbb{E}_{A_v \mid s}[H(Y \mid s, v = A_v)]. \tag{13}$$

For a compatible joint law this is nonnegative conditional mutual information, distinct from geometric displacement μ_v in (10). Refinement can be tested by revealing held-out archival answers.

B. Phenotypic construction: querying a reflexive system

Under construction, response A_t to query q may alter latent state Z_t:

$$K_q(a, z' \mid z) = O_q(a \mid z)\, T_q(z' \mid z, a). \tag{14}$$

TABLE I

GSS 2024 left/right reference poles. The 27 listed item–response assignments are the complete nonmissing coordinates of the two pole states used to compute I_Φ. All other survey variables are unspecified. Topics paraphrase item wording; categorical responses match the simulation definitions. The poles yield $I_\Phi = -1$ and $+1$.

Item	Survey topic	Left	Right
abany	Abortion on request	yes	no
abdefect	Abortion: fetal defect	yes	no
abnomore	Abortion: no more children	yes	no
abpoor	Abortion: low income	yes	no
abrape	Abortion after rape	yes	no
absingle	Abortion: unmarried	yes	no
bible	Bible interpretation	ancient book	inspired word
colcom	Communist as college teacher	not fired	yes, fired
confed	Trust in federal government	a great deal	hardly any
conlabor	Trust in organized labor	a great deal	hardly any
grass	Marijuana legalization	should be legal	should not be legal
gunlaw	Permit to purchase gun	favor	oppose
libcom	Communist books in library	not remove	remove
libmslm	Muslim cleric books in library	not remove	remove
natarms	Military spending	too much	about right
natenvir	Environmental spending	too little	about right
natfare	Welfare spending	too little	too much
natsoc	Social Security spending	too little	about right
owngun	Guns in household	no	yes
pillok	Contraception access	strongly agree	disagree
polabuse	Police strike insulting citizen	yes	no
pray	Frequency of prayer	never	several times a day
prayer	Bible prayer in public schools	approve	disapprove
religten	Personal religiosity	no religion	strong
rowngun	Respondent owns a gun	no	yes
shotgun	Owns a shotgun	no	yes
spkcom	Communist public speech	yes, allowed to speak	not allowed

Passive observation has $T_q(z' \mid z, a) = \delta_z(z')$; otherwise transitions depend on prior beliefs, framing, trust, or history. Cross-sectional LSM conditionals cannot identify T_q, even with perfect observational fit.

Clarification, temporary salience, and lasting revision call for independent outcomes, delayed measurement, and no-query controls. We test whether statistical geometry predicts responsiveness as well as reconstruction.

C. Why question-order effects are not automatically persuasion

A coherent static joint model implies

$$P(a \mid s, v)P(b \mid s, v = a, w) = P(b \mid s, w)P(a \mid s, w = b, v). \tag{15}$$

With a coherent joint law, terminal predictions after both answers are order-invariant, though intermediate states differ. Incompatible local conditionals can create order artifacts (15) [22]; genuine human operators K_A, K_B may fail to commute. Simulation alone cannot distinguish them.

VI. Native GSS 2024 Computational Experiments

A. Adaptive questioning of fixed synthetic personas

We used six fixed synthetic GSS 2024 states (Table II), querying survey items directly without running LLM question mapping or verbalization. The native optimizer maximized expected rightward drift, minimized it for leftward drift, minimized absolute drift (neutral), or sampled eligible items randomly. Answers were drawn from LSM conditionals. At $T = 20$ and 40 repetitions per persona/policy, $6 \times 4 \times 40 = 960$ trajectories contain 19,200 draws.

Right- and left-seeking policies moved the index in their expected directions across all six personas; neutral minimized *geometric drift*, not psychological sensitivity.

B. Question selection as observed paths

Each observed question was assigned a shared categorical index without reordering runs or constructing modal paths. Of 228 items across all policies, 161 appeared in the 18 displayed right/left/neutral heatmaps (six personas by three policies; 40 runs of 20 questions). The global index permutation approximately minimizes

$$\mathcal{C}(\pi) = \sum_{p,u,t} \sum_{r<r'} \left|\pi(v_{purt}) - \pi(v_{pur't})\right|, \tag{16}$$

Here u ranges over displayed policies. The all-pairs objective is independent of run numbering. The 67 random-only items remain in the lookup; index distances have no scientific meaning.

C. Order sensitivity with twenty-question content fixed

For each persona and right/left/neutral policy, we froze the 20 items from replicate zero without endpoint selection. Each of the 18 sets was replayed in original, reversed, and two fixed shuffled orders, with 40 conditional-sampling repetitions: $18 \times 4 \times 40 = 2{,}880$ trajectories. Paired orders shared question-specific uniform draws to reduce Monte Carlo noise. We compared final-sign and movement-direction reversals:

$$F^{\text{end}}_{pSr\pi} = \mathbf{1}\{I^{(\pi)}_{20} I^{(0)}_{20} < 0\}, \tag{17}$$

$$F^{\text{move}}_{pSr\pi} = \mathbf{1}\{(I^{(\pi)}_{20} - I_0)(I^{(0)}_{20} - I_0) < 0\}. \tag{18}$$

Sign reversals require both signed quantities to exceed 10^{-9} in magnitude.

Figure 2 shows paired mean contrasts. Among 720 matched comparisons per source policy, movement-direction reversals numbered 107 (right), 68 (left), and 10 (neutral); final-sign reversals numbered 69, 54, and 0. These coupling-dependent counts are descriptive, especially near zero drift. All 54 pointwise bootstrap intervals for mean contrasts contained zero.

Coupled pathwise reversals do not necessarily change marginal terminal distributions: a coherent joint law can produce different paired paths with identical terminal laws. Full endpoint distributions require separate comparison.

TABLE II

Six pinned synthetic personas in the native GSS 2024 simulations. Persona descriptions are reproduced from `gss2024_spectrum.yaml`. The initial pole-relative ideology index $I_\Phi(s_0)$ comes from the 20-question, 40-repetition experiment's `initial_I` field in `adaptive/endpoints.csv`; it is not the persona's self-reported political ideology. Positive values point toward the designated right pole and negative values toward the left pole.

Persona	Pinned description	Initial $I_\Phi(s_0)$
Alabama	45-year-old white man living in open country outside a small town in rural Alabama, in the South. He is married and has two children. He works full time in farming. He is a veteran who served 2 to 4 years of active duty in the armed forces. He reads a newspaper every day. His education ended with a high school diploma. He thinks of himself as conservative and is a strong Republican. He is Protestant and attends religious services every week.	0.075706
New York	22-year-old white woman living in New York City, a large central city in the Northeast. She has never been married and has no children. She works full time in retail. She has never served in the armed forces. She reads a newspaper every day. She has a bachelor's degree. She thinks of herself as extremely liberal and is a strong Democrat. She has no religion and never attends religious services.	−0.163807
Texas	61-year-old white woman in a small town in Texas, in the South. She is married with three children and keeps house. Her education ended with high school. She thinks of herself as extremely conservative and is a strong Republican. She is Protestant and attends religious services several times a week. She reads a newspaper every day.	0.045379
Seattle	34-year-old white man living in Seattle, a large central city in the West. He has never been married and has no children. He works full time and has a graduate degree. He thinks of himself as liberal and is a Democrat, but not a very strong one. He has no religion and never attends religious services. He reads a newspaper a few times a week.	−0.119639
Arizona	39-year-old Mexican American man living in a medium-size city in Arizona, in the West. He is married with three children and works full time. His education ended with high school. He thinks of himself as conservative and is a Republican, but not a very strong one. He is Catholic and attends religious services nearly every week.	−0.020437
Atlanta	47-year-old Black woman living in Atlanta, a large central city in the South. She is divorced with two children and works full time. She has a bachelor's degree. She thinks of herself as liberal and is a strong Democrat. She is Protestant and attends religious services every week. She reads a newspaper every day.	−0.030807

a) Cross-environment limitation.: Historical ten-question local and CI runs yielded different initial indices despite matching persona, semantic-map, and pole-file hashes. Model binaries were not hashed in both manifests, so absolute cross-environment comparison requires runtime/model parity checks. Within-study results remain consistent.

VII. Testable Hypotheses and Future Human Experiments

H1 (predictive refinement). Given an initial answer set s_0, DTAG's conditional distributions for unasked questions predict participants' actual responses. Held-out log score should improve as additional answers enter s_t, while probability forecasts remain calibrated; demographic-only and fixed-query baselines provide comparisons.

H2 (directional adaptive querying). After a common initial question sequence estimates baseline $I_\Phi(s_0)$, the one-step rightward/leftward policies based on $\mu_v(s)$ should produce corresponding directional changes in the index computed from participants' subsequent *actual* answers, relative to neutral and random-query controls. Separately held-out responses assess whether any change extends beyond the queried items.

H3 (question-order effects). After the same baseline, presenting an identical question set in forward versus reversed order should alter the resulting ideological response, measured on independent held-out items. The predicted direction and size of the order contrast must be specified before observing outcomes; intermediate index movement or coupled synthetic replay differences alone are insufficient.

a) Future validation on Prolific.: We propose, but have not conducted, randomized human experiments. A common baseline battery establishes s_0. For H1, distributions are recorded *before* each response and evaluated over increasing anchor budgets. For H2, participants are randomly assigned to rightward, leftward, neutral, or random question-selection policies without persuasive messaging. For H3, a separate randomization assigns the same prespecified modules in orders AB or BA. Independent outcome items Y, optional 7–14-day follow-up, preregistered contrasts, and a pilot-based power calculation will distinguish model-index refinement from persistent response change. Prolific supports longitudinal recontact [24]. The delayed order contrast is

$$\tau_{\text{order}} = \mathbb{E}[Y_{\text{delay}} \mid \text{do}(AB)] - \mathbb{E}[Y_{\text{delay}} \mid \text{do}(BA)]. \quad (19)$$

The native LSM does not identify causal belief transitions; this is future validation, not a finding. Human-subjects oversight and safeguards against individually targeted political influence will be required. The focus is question elicitation, not social-media exposure as in [11].

VIII. Discussion and Limitations

DTAG models *what a respondent might report*, not yet *what an interaction makes them believe*. Associations can reflect common causes, sorting, survey design, or history.

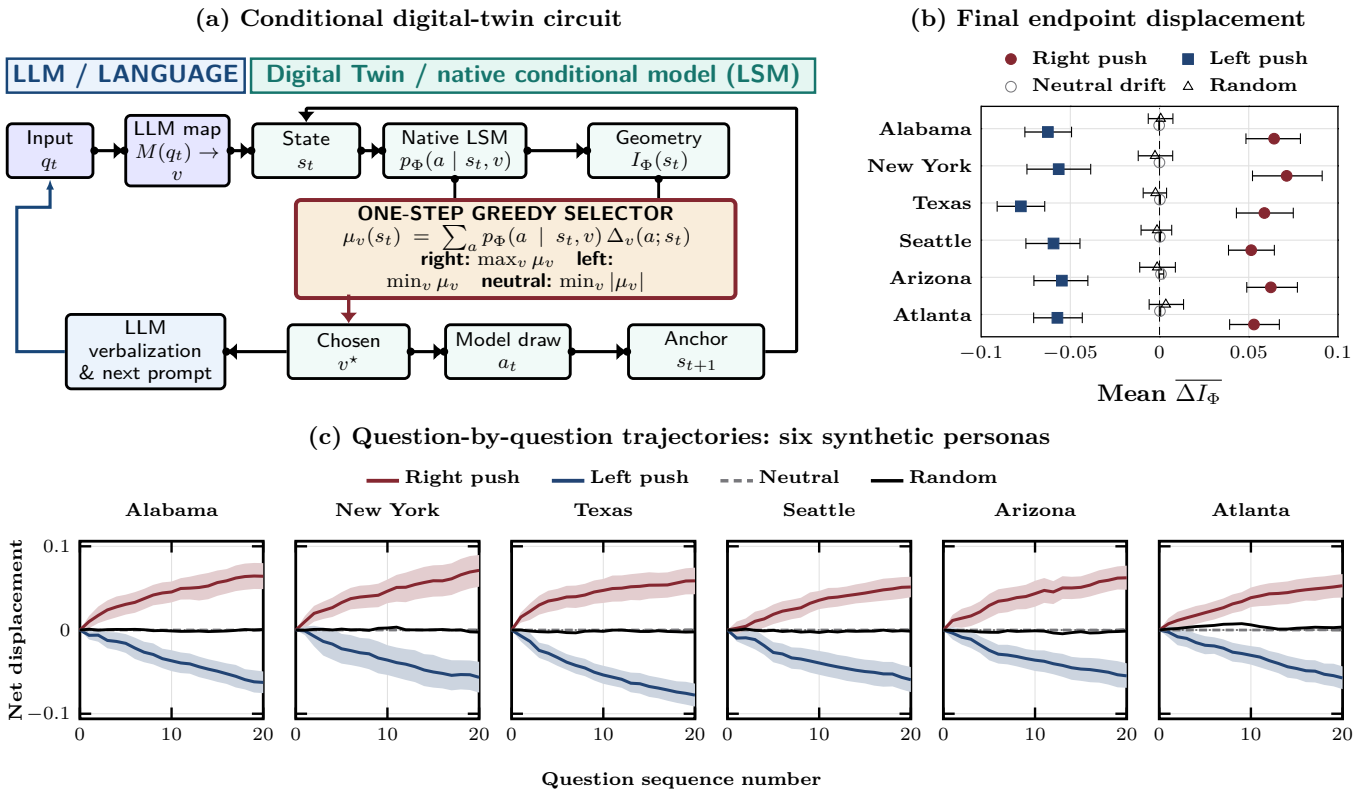


Fig. 1. **DTAG digital-twin circuit and adaptive native GSS 2024 results.** (a) A semantic LLM interface maps and verbalizes categorical survey questions, while the conditional LSM inside the digital twin predicts unasked responses; a myopic greedy circuit selects the next question using expected one-step pole-relative displacement and updates the anchored state with a model draw. (b) Mean displacement after 20 questions across six synthetic personas; large markers and horizontal whiskers show 95% bootstrap uncertainty. (c) Six aligned trajectories from 40 repetitions per persona and policy; shading denotes pointwise 95% bootstrap intervals. Right = dark red, left = dark blue, neutral = dashed gray, random = black. These are native conditional-model trajectories, not observed human persuasion.

Incompatible LSM conditionals or semantic mappings may create synthetic order effects unrelated to human reflexivity.

Under passive observation of a coherent joint law, the posterior mean of a fixed trait $g(X)$ is a martingale. DTAG's nonlinear index $I_\Phi(s)$ is generally not that posterior mean and may drift without persuasion. Limitations also include pole and wave dependence, missingness calibration, wording, demand, response modes, and eviction of old anchors. Human experiments need measurement invariance and delayed outcomes.

IX. Conclusion

DTAG uses high-dimensional conditional predictions for adaptive querying and model-based phenotypic refinement. Synthetic trajectories need not represent human change; held-out reconstruction and randomized human order experiments remain distinct validation tests.

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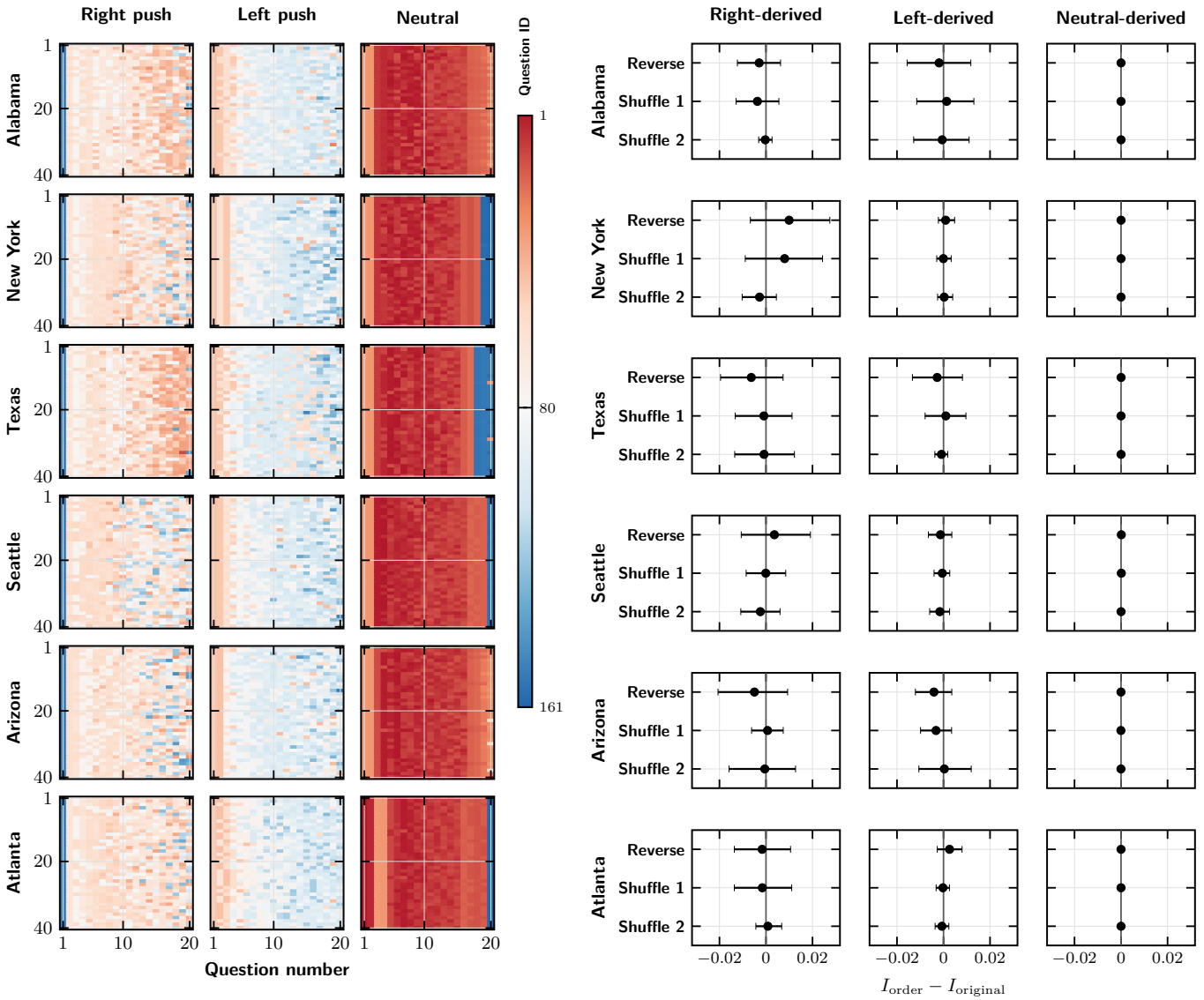


Fig. 2. **Question identities and fixed-item order effects over all six personas.** Left: 18 nearly square categorical heatmaps show 40 independent runs over 20 question positions. The single colorbar denotes optimized question IDs (blue: low; red: high), not signed ideological effects. A single globally optimized categorical map covers 161 displayed items from 228 observed questions. Right: original order versus reversal and two shuffles of fixed 20-item sets, displaying paired endpoint differences with 95% bootstrap confidence intervals. The items are fixed; the displayed contrasts are model-based.

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